



CAREER OBJECTIVE

- Seeking a role in Embedded Systems Hardware Design where I can apply my skills to drive innovation and contribute to both organizational and personal growth.

Professional Synopsis

- Strong foundation in electronics fundamentals and circuit analysis, with extensive experience in hardware design, board-level testing, and troubleshooting.
- Hands-on experience in Mixed signal hardware design including communication protocols, Microprocessors, Microcontrollers, FPGA, ADC, DAC, Op-amp, sensors and IO's.
- Skilled in schematic design using ALTIUM Schematic design, layout Design, and evaluating prototypes from engineering samples to mass production.
- Expertise in interpreting hardware datasheets and implementing designs based on those specifications.
- Knowledgeable in peripheral interfaces such as SPI, UART (RS422 & RS232), and I2C, with hands-on testing experience using tools like Hercules, Tera Term.
- Extensive hands-on experience in hardware Design & testing, debugging, and preparing quality assurance plan documents in line with user/customer specifications and quality guidelines.
- Hardware architecture design, power budgeting, time budgeting according to requirements
- Solid understanding of DC-DC & AC-DC converters, LDOs, and amplifiers.
- Experienced in preparing technical documentation for the fabrication and testing of electronic units.
- Competent in the use of oscilloscopes, signal generators, power analyzers, and other test equipment.
- **Component Selection** and evaluation, Circuit simulation using **LTSpice** and TINA

Academic Profile

- B. Tech (Electronics and Communication Engineer) from VITAM College of Engineering, Visakhapatnam-in 2011.
- Diploma (Electronics and Communication Engineer) from T.K.R Polytechnic college in 2007.
- SSC from Board of Secondary Education in 2003.

Employment Details

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| <ul style="list-style-type: none">• Sr Hardware Design Engineer– R&D Department
pathnsitu biotechnologies Pvt. Ltd,
Hyderabad
<i>May 2023 – Present</i>• Engineer – R&D Department• Astra microwave products Ltd., Hyderabad
<i>June 2018 – April 2023</i> | <ul style="list-style-type: none">• RF TESTING Engineer
DLRL, Hyderabad
<i>August 2016 – May 2018</i>• Hardware & PCB Design Engineer• Sri lighting Innovations Pvt . Ltd, Hyderabad
<i>May 2014 – July 2016</i>• Hardware & PCB Design Engineer• Eco Led Illumination Pvt. Ltd., <i>March 2011– April 2014</i> |
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Project #1: Pathnsitu biotechnologies Pvt. Ltd

- Project title : Tissue Flotation Bath (With Bowl & WITH Out bowl)
- Client : Internal use

Role: Hardware Design Engineer

Description:

- Designed a tissue flotation bath system to aid in the precise and controlled flotation of tissue samples.
- Responsible for schematic design, component selection, and PCB layout for the bath's electronics.
- Ensured high accuracy in temperature control, fluid circulation, and user interface design for optimal results.

Responsibilities:

- Requirement study and low-level requirements derivation.
- Architecture design, schematic capture.
- Design calculations, component selection.
- PCB Design & review.

Project #2: Pathnsitu biotechnologies Pvt. Ltd

- Project title : Tissue Embedding Center
- Client : Internal use

Role: Hardware Design Engineer

Description:

- and development of a **Tissue Embedding Center**, a critical tool used in the preparation of tissue samples for microscopic analysis.
- Designed the hardware for embedding modules, focusing on temperature control, sample manipulation, and user interface integration.

Responsibilities:

- Schematic Design: Created detailed electronic schematics for the tissue embedding center, including temperature control circuits, motorized component control, and sensor integration for accurate sample handling.
- Component Selection: Responsible for selecting and integrating components like temperature sensors, heating elements
- System Integration: Collaborated with mechanical and software engineers to ensure smooth integration of the hardware with mechanical systems and software control systems.
- Testing and Validation: Conducted extensive testing to validate the hardware design, ensuring proper functioning of temperature regulation and system durability.
- Documentation: Developed comprehensive documentation for hardware setup, troubleshooting, and calibration procedures.

Project #3: Pathnsitu biotechnologies Pvt. Ltd

- Project title : Single Arm Linear H&D Project

- Client : Internal use

Role: Hardware Design Engineer

Description:

- Designed and developed the hardware for a Single Arm Linear System, focusing on precision movement, motor control.
- The project included the development of a linear actuator system, leveraging hardware and design expertise to enable efficient, controlled motion for precise tasks.

Responsibilities:

- Architecture design, schematic capture, design calculations.
- Component selection, PCB review.
- Test plan preparation, testing.
- Customer interaction, design modifications for the issues faced.
- Technology Transfer to Production.

- **PHH (PASSIVE HOMMING HEAD) Astra Microwave Products LTD.**

CLIENT: DRDL HYD

Responsibilities:

- Preparing the wiring details of all modules - INTGRATION SYSTEM.
- Testing of RF components at bench level
- RF cable testing and Documentation

- **Drone Control system- Astra Microwave Products LTD.**

CLIENT: DRDL HYD

Responsibilities:

- Preparing the wiring details of all modules - INTGRATION SYSTEM
- Testing of RF components at bench level
- RF cable testing and Documentation
- Low noise amplifiers
- Power amplifiers
- Mixers
- Power dividers.

PROFESSIONAL ABILITIES

- Active team player with commendable communication skills.
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- Good listener and quick learner with strong interpersonal and analytical skills.
- Ability to manage time effectively and efficiently through scheduling and prioritizing.
- Strong minded and consistent in results.

Personal Details

- Father's Name : M Vijaya
- Mother's Name : Smt. M.Nirmala
- Date of Birth : 15-06-1987
- Marital Status : Married
- Sex : Male
- Languages Known : English, Hindi, Telugu
- Address : 6-34,Vemanda post, Unguturu mandal, Krishna Dist, Andhra Pradesh-521311.

Declaration:

I hereby declare that the information furnished above is true to the best of my Knowledge and belief.

Date:

Place:

(M. Pavan Kumar)